

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Consider the vector space $V = \mathbb{C}^3$ with the scalar field $\mathbb{C}$ and addition and scalar multiplication defined as follows:

$$(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 + 7, y_1 + y_2, z_1 + z_2 - 4)$$

for any $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ and

$$\alpha(x_1, y_1, z_1) = (\alpha x_1 + 7\alpha - 7, \alpha y_1, \alpha z_1 - 4\alpha + 4)$$

for any $\alpha \in \mathbb{C}$. This defines a vector space. (You do not have to prove this!)

- (a) Find the zero vector for V .
 - (b) Find the additive inverse of (x, y, z) in V .
2. (NS 1.2.12b) Let U and W be subspaces of a vector space V . Prove that $U \cup W$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$.
 3. Let $V = \mathbb{C}^{n \times n}$ for $n \geq 2$ be the complex vector space with the usual matrix addition and scalar multiplication. Let $U_1 = \{A \in \mathbb{C}^{n \times n} : A^T = A\}$ be the set of symmetric matrices in V and let $U_2 = \{A \in \mathbb{C}^{n \times n} : A^* = A\}$ be the set of hermitian matrices in V . One of U_1 and U_2 is a subspace of V while the other is not.

Determine which is which and give full justification for each independently (i.e., “this one is a subspace because the other is not” or vice versa is not valid).

4. Find a basis for the real vector space

$$V = \{(a, b, c, d) \in \mathbb{R}^4 : c = -6b, d = 2a\}$$

with the usual addition and scalar multiplication. Make sure to prove your proposed basis is, in fact, a basis!

5. Under what conditions on α is

$$\{(1, \alpha, 0), (\alpha, 0, 1), (1 - \alpha, \alpha, 1)\}$$

a basis for $\mathbb{R}^3$?



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 1.1: 2, 8, 9
- Section 1.2: 2, 3, 4, 8
- Section 1.3: 1, 3, 6, 9, 12
- Section 1.4: 1, 5, 7, 8, 13
- Section 1.5: 1ab, 2, 7, 10, 11

1. Let U be a subspace of V . Show that $\dim(U) \leq \dim(V)$.

